TAKE-HOME ASSIGNMENT

Person Counting in RGB and Thermal Drone Images

AI-based prototype for detecting objects classified as people and estimating the total number of people in aerial imagery.

CANDIDATE	CANDIDATE PROFILE	EXPECTED EFFORT
Stav	**Software Engineering Graduate**	**Approximately 30 Hours**

ASSIGNMENT OVERVIEW

Background and Objective

Drones capture pairs of images of the same or a similar area using a standard RGB camera and a thermal camera. The images may include people at different scales, dense groups, partial occlusions, changing lighting conditions, and differences in resolution, angle, and field of view between the two sensors.

> **Primary objective:** build a working AI prototype that determines whether visible objects are people and counts the total number of people in each image.

The objective is **not** to identify specific individuals, distinguish one person from another, perform face recognition, or track personal identity. Location outputs such as bounding boxes or points should be used only to validate the result, visualize detections, and reduce duplicate counting.

What the assignment evaluates

Engineering capability

- Building a clear and reproducible computer-vision pipeline
- Organizing code, configuration, dependencies, and outputs
- Testing, debugging, and handling common failure cases

Technical judgment

- Researching and selecting appropriate models
- Understanding metrics, thresholds, and limitations
- Explaining and defending technical decisions

Expected scope

The expected effort is approximately 30 working hours. The assignment is designed as an engineering prototype rather than a production system or a research project. The candidate should prioritize a reliable baseline, meaningful validation, clean code, and clear conclusions.

> **Scope management:** if not all planned work is completed, document what was completed, what remains, why the work was prioritized in that way, and what the next engineering steps would be.

Provided data

Several RGB and thermal image pairs will be supplied. The following pair illustrates the expected input format.

RGB image example. Standard visual imagery captured from a drone.

Thermal image example. Thermal view of the same or a similar scene.

CORE REQUIREMENTS

Mandatory Work

1. Data analysis

Review the provided image pairs and briefly describe the challenges that affect person counting. Address the most relevant issues, such as:

- People appearing very small relative to image size
- Dense groups, partial occlusions, and overlapping detections
- Differences between RGB and thermal imagery
- Differences in resolution, viewing angle, and field of view
- Lighting conditions, heat sources, background clutter, and objects that may resemble people
- Possible timing differences between paired images

Explain how these factors influence model choice, input resolution, preprocessing, confidence thresholds, and validation.

2. Model research and selection

Review at least two relevant models or technical approaches. A written comparison is required, but only one primary implementation is required.

The comparison should address:

- Suitability for small-object detection in aerial imagery
- Suitability for RGB and thermal input
- Availability and relevance of pretrained weights

- Inference speed and hardware requirements
- Support for fine-tuning
- Model and code licensing
- Potential future use in cloud or edge environments

Select the final approach and explain its advantages, disadvantages, and expected limitations.

3. Baseline implementation

Implement an end-to-end pipeline that supports both RGB and thermal images and includes:

1. Loading a single image or a directory of images
2. Identifying image modality, through metadata, naming convention, configuration, or explicit input
3. Applying appropriate preprocessing
4. Running the selected model
5. Filtering outputs to objects classified as people
6. Applying confidence, IoU, NMS, or other relevant filtering
7. Counting the final detections
8. Saving an annotated image for visual review
9. Saving structured output in JSON or CSV format

```json
{
  "image_name": "DJI_example.jpg",
  "modality": "rgb",
  "people_count": 18,
  "detections": [
    {
      "bbox": [120, 85, 170, 210],
      "confidence": 0.91,
      "class": "person"
    }
  ]
}
```

4. Ground truth and evaluation sample

Create a small, representative evaluation sample from the provided data. It should include:

- Several RGB and thermal images or image pairs
- Manually verified person counts for each selected image
- Annotations that allow detection-level evaluation, such as bounding boxes or points
- Documentation of the annotation format and assumptions

A large annotation project is not expected. Select enough examples to support a meaningful initial evaluation.

VALIDATION AND ANALYSIS

Performance Evaluation

5. Required metrics

Evaluate RGB and thermal results separately. At minimum, report:

- Ground-truth person count for each image
- Predicted person count for each image
- Absolute counting error per image
- Mean Absolute Error (MAE)
- Precision
- Recall
- Examples of false positives and false negatives

Where practical, also report F1 score, mAP@0.5, average inference time, or performance by person size. These additional metrics are useful but should not replace a clear counting analysis.

6. Result analysis

Include a short analysis covering:

- Cases in which the model performed well
- Cases in which the count was incorrect
- Likely causes of false positives and false negatives
- The effect of confidence, input resolution, IoU, and NMS settings
- Main differences between RGB and thermal performance

7. RGB and thermal comparison

Compare the two modalities and answer the following:

- Which modality produced more accurate counts on the evaluation sample?
- Under which conditions was thermal imagery more effective?
- Under which conditions was RGB imagery more effective?
- What were the main error sources in each modality?
- Could one modality help validate detections from the other?

> **Important:** a full multimodal fusion system is not required. A clear comparison and a practical proposal for future fusion are sufficient for the mandatory scope.

8. Code quality and reproducibility

The submitted repository should demonstrate software-engineering discipline appropriate for a graduate-level candidate:

- Clear project structure and separation of responsibilities
- Readable names, comments where useful, and limited duplication
- Configuration rather than hard-coded paths and thresholds
- Basic input validation and error handling
- Reproducible installation and execution instructions
- Reasonable logging and deterministic behavior where applicable
- At least a small set of meaningful tests for core logic

OPTIONAL EXTENSIONS

Additional Work, When Justified

OPTIONAL

Fine-tuning experiment

Perform a limited fine-tuning experiment only if the available data and baseline results justify it.

Document the initial weights, dataset split, input size, batch size, learning rate, epochs, augmentations, checkpoint selection, and overfitting controls.

OPTIONAL

Multimodal result fusion

Propose or implement a basic method for matching or validating detections between RGB and thermal images.

Examples include box matching, geometric alignment, score fusion, or using one modality to confirm the other.

OPTIONAL

Deployment packaging

Provide a Dockerfile, benchmark an edge-oriented model, or describe a cloud deployment architecture.

OPTIONAL

Advanced evaluation

Add mAP@0.5:0.95, size-based analysis, resource usage, or evaluation across additional public datasets.

Fine-tuning is not mandatory. A technically sound decision not to fine-tune is acceptable when supported by evidence, such as limited data, insufficient annotation quality, or satisfactory pretrained performance. In that case, explain how fine-tuning should be performed in future work.

Suggested time allocation

Activity	Suggested Time
Data review and problem definition	3 hours
Model research and technical selection	4 hours
Baseline implementation	8 hours
Ground truth and evaluation sample	4 hours
Validation and result analysis	5 hours
Optional experiment or deeper analysis	2 hours
Testing, cleanup, documentation, and presentation	4 hours
Total expected effort	**30 hours**

This allocation is guidance only. Candidates may adjust it according to their chosen approach and findings.

SUBMISSION

Required Deliverables

1. Source-code repository

- Inference pipeline for RGB and thermal images
- Preprocessing and post-processing code
- Counting and structured-output generation
- Evaluation scripts
- Tests for selected core components
- `requirements.txt`, environment file, or equivalent dependency specification
- Fine-tuning code only if fine-tuning was performed

2. README

- Problem definition and solution overview
- Installation and execution instructions
- Models considered and reasons for the final choice
- Preprocessing, post-processing, and key parameters
- Annotation and validation methodology
- Results and RGB-versus-thermal comparison
- Known limitations and recommended next steps
- Disclosure of significant AI-assisted work

3. Results package

- Annotated RGB and thermal images
- Ground-truth and predicted counts
- JSON or CSV result files
- Metric summary and comparison tables
- Examples of correct detections and errors

4. Technical presentation

Prepare a presentation or live walkthrough of approximately 15 minutes covering the problem, model selection, implementation, validation, results, limitations, and recommended next steps.

Technical review expectations

During the review, the candidate may be asked to:

- Explain selected parts of the code and data flow
- Run the solution on an additional image
- Modify a threshold, parameter, or configuration value

- Diagnose an error or unexpected output
- Explain how a code change affects the final count
- Discuss how the solution could be improved or productionized

AI USAGE AND EVALUATION

Use of AI Tools and Assessment Criteria

AI tools are permitted. The use of LLMs, coding assistants, AI agents, pretrained models, and open-source solutions is allowed and encouraged when appropriate.

The candidate remains fully responsible for the submitted solution, including:

- Correctness and reliability of the implementation
- Code quality, structure, maintainability, and security considerations
- Selection and understanding of libraries, models, and dependencies
- Reproducibility of the environment and results
- Quality of tests, validation, and technical conclusions
- Compliance with the licenses of models, code, and datasets used

During the technical review, the candidate is expected to demonstrate ownership of the solution. This includes explaining the implementation, identifying limitations, debugging issues, and making changes to the code or configuration without relying entirely on generated answers.

Significant use of AI-generated code, analysis, or documentation should be disclosed in the README, together with a short explanation of how the output was reviewed and validated.

Evaluation criteria

Evaluation Area	Weight
Understanding of the problem and data analysis	**15%**
Model research, selection, and technical reasoning	**20%**
Implementation, software design, and code quality	**25%**
Validation, counting metrics, and result analysis	**20%**
RGB and thermal comparison	**5%**
Documentation, presentation, and command of the solution	**15%**